

200 series

SH-WMC2000

Multi-parameter

water quality controller

User manual

Reading Instructions

Information for Users

Thank you very much for choosing to use our SH-WMC2000 multi-parameter water quality controller (hereinafter referred to as “WMC2000 controller”), to manage local data acquisition, system configuration and operational management of 200 series sensors, probes and measurement devices for water quality control. Before using this product, please read this user manual carefully. This manual covers important information and data for the use of the product, which must be strictly observed by the user to ensure its normal operation. The information will help you to use the product correctly and to obtain reliable data control and long-term operation.

General description

This manual provides a detailed description of the installation, commissioning, operation and maintenance of the WMC2000 controller, as well as describing its functionalities, instrument composition and configuration parameters.

This manual instructs the user on the proper installation and operation of the WMC2000 controller, as well as preventive maintenance work on the controller, to guarantee its continuous and reliable operation.

The product described in this manual has been rigorously inspected before leaving the factory to ensure top quality. In order to ensure safe, high-quality operation and to obtain correct analysis results, the user must strictly follow the instructions described in this manual.

This manual provides all the information required for the proper use of the WMC2000 controller. It provides an accurate reference for use by technicians with specialized training or knowledge of the instrument's operational controls. Before operating the instrument, make sure that the safety and warning messages mentioned in this manual are correctly understood and applied to the actual operation.

Safety and warnings

Unregulated operation under any circumstances may cause injury to personnel and instruments, please observe all safety codes for electrical appliances and equipment.

Make sure the power supply is disconnected before servicing or disassembling electrical components.

Please make sure all external wiring is correct before switching on the instrument, any short-circuiting may result in loss of data or destruction of the instrument's programming.

Delivery and transport

Shipment requirements are in accordance with the terms of the ordering contract.

Please read the appropriate information on the packaging material carefully when opening the box to ensure that the opened goods are complete and undamaged. Please keep the outer packaging of the product for use when the instrument needs to be returned for repair.

Warranty and maintenance

Specific warranty and maintenance requirements are in accordance with the terms and conditions of the order contract.

During the warranty period and in line with the warranty scope, free maintenance services will be provided, mainly including product maintenance under warranty, spare parts repair and replacement and remote technical support.

Failures that exceed the warranty period or occur within the warranty period as follows are out-of-warranty repairs and do not provide free warranty service, failures include but are not limited to:

- (1) Due to improper use (water ingress, corrosion, fire, strong electricity strung in, etc.);
- 2) Damage caused by force majeure (earthquakes, lightning strikes, floods, etc.);
- (3) Unauthorized alterations within the product without permission;
- (4) Failure to use the product in accordance with the user's manual and training provisions, causing damage to the product.

Regarding the products developed and manufactured by our company, we strictly comply with the relevant national regulations on the disposal of used products.

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1 Instrument overview

1.1 Instrument Introduction

The WMC2000 controller can be connected with all kinds of 200 series water quality sensors, probes and measurement devices, to display the measured values and electrode status in real time. The controller can calibrate each device, store and transmit measurement data and calibration information and configure each device. The interface is simple and easy to operate, and can be used quickly by non-professionals. Connectable sensors and probes include: pH, conductivity, dissolved oxygen, turbidity, ion-selective electrode ammonia nitrogen, COD, ORP and other measurement equipment. The controller can be connected to 4 devices at the same time.

1.2 Product features

The main technical parameters of the controller are as follows:

Product name	Multi-parameter water quality controller
Display screen	7-inch touch screen
Display size	(154 x 86) mm
Display resolution	800 x 480 pixels
Digital communications	2-way RS485 communication, standard Modbus protocol
Working environment	(-10~60) °C, (0~95) %RH
Protection class	IP65
Dimensions	(235*185*117) mm
Operating voltage	110-220 Vac, 50-60 Hz

2 Controller installation

2.1 Instrument environment

The environment in which the controller is installed and used needs to meet the following requirements:

- avoid damage caused by external impacts, cuts and pressure during installation.
- the installation environment needs to be protected from moisture and direct sunlight.
- the installation should be firmly fixed during the installation, to prevent its possible damage in case of falling.

2.2 Instrument wiring

Controller wiring, see Figure 2.1: please open the external protective cover, on the right is the Vac power supply interface, the wire connections as following:

- the brown wire for the L
- the blue wire for the N
- the yellow-green wire for the PE (ground)

that is connected to the interface with the circuit board.

On the left there are five Modbus 485 connections as following:

- PORT1: connection to the sensors, probes and measurement devices, max. 4 units connected in parallel
- PORT2: not used
- PORT3: relay digital outputs modules
- PORT4: 4-20 mA analog outputs modules
- PORT5: interface for external communication, the detailed external transmission protocol is shown in section 5.

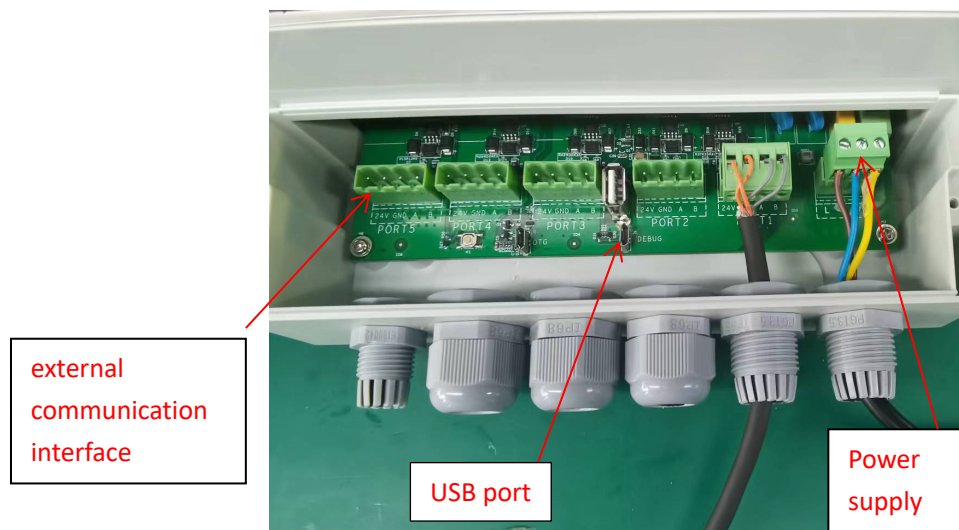


Figure 2.1 Instrument Wiring

Between PORT 2 and PORT 3 there is the USB port, used for SW update and data export; wiring schematics is reported in the following Figure 2.2.

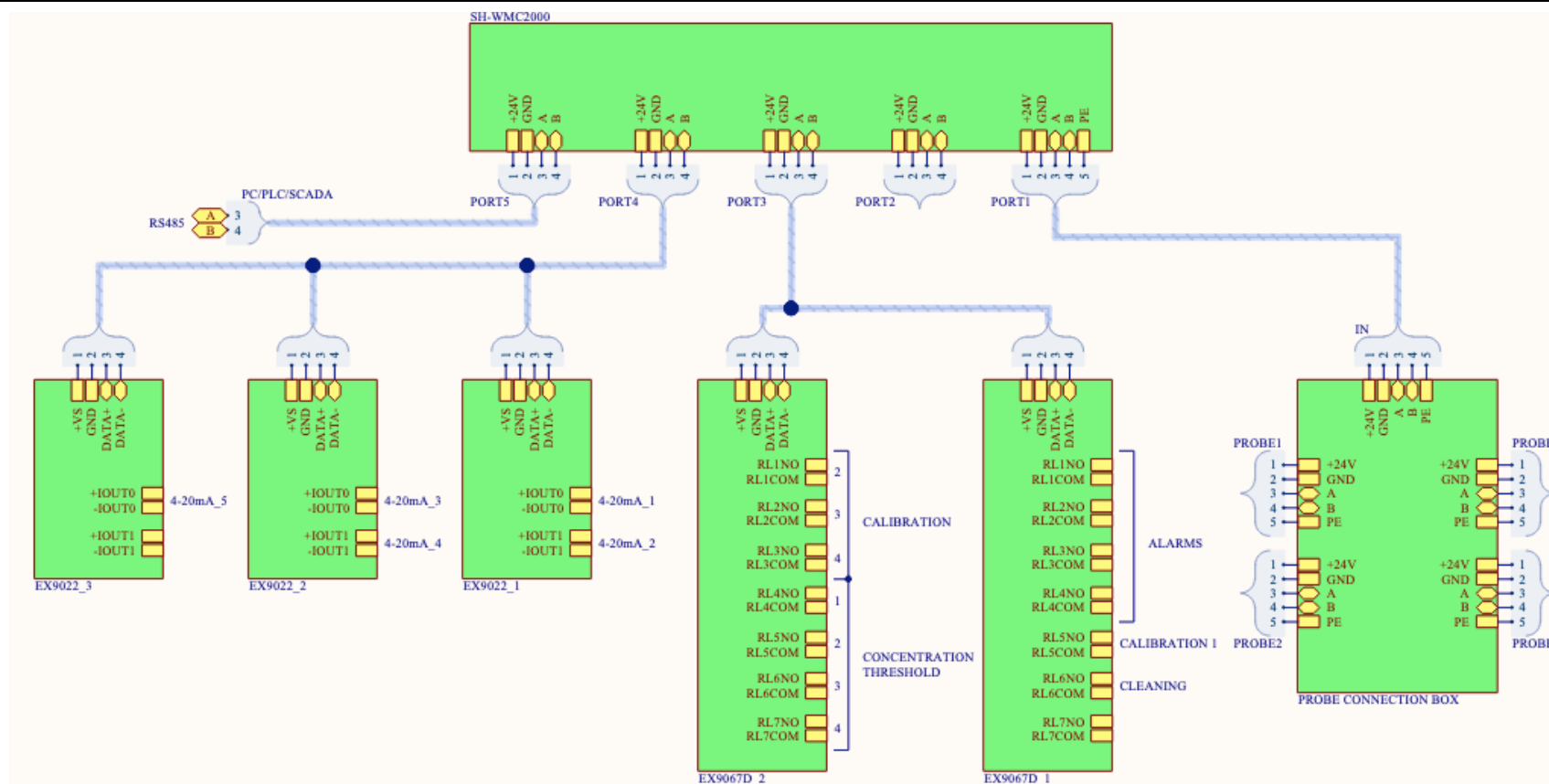


Figure 2.2 Wiring schematics

2.3 Probe connection box (optional)

In case of multiparametric connection of two or more measurement devices to PORT1, the probe connection box is available on request, see Figure 2.3.

Two devices will be connected by their own cables through the left side glands, the other two on the right side glands.



Figure 2.3 Probe connection box - external

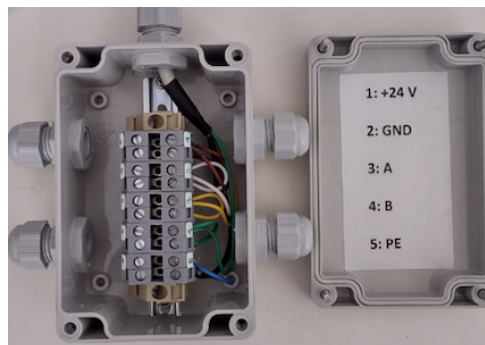


Figure 2.4 Probe connection box – internal wiring connections

Each wire of the device cables have to be connected according to the pin schema shown here above.

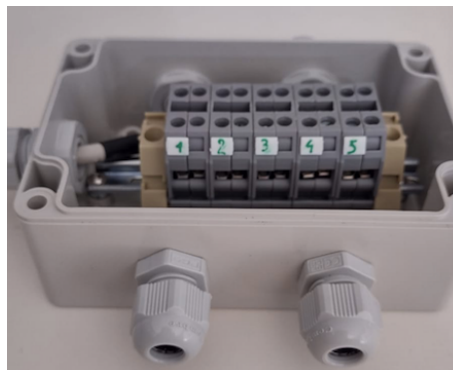


Figure 2.4 Probe connection box – internal wiring connections

For each signal type (1 ..5), two wires have to be connected on the left clamp and the other two on the right clamp.

3 Equipment operation

3.1 Main interface functions

After the controller is powered on, the main interface window is shown; on the upper side the product model and software version are displayed. On the left of the company logo there is an icon, to be used for user login and querying the system information of the controller. At the right side of the display there are the buttons for the selection of the operations of the connected probes. The central part of the interface displays the measurement data of each connected device, in the form of a four-cell grid and the corresponding relationship of each parameter. The temperature measured by each device is displayed on the lower right side of each parameter's section, while the eventual second measured parameter on the lower left side. Take the five-parameter configuration in the figure below as an example, the first row is the parameter name, the second row is the measurement value with the unit, and the third row is the test value of the auxiliary parameter detected by a small number of devices simultaneously. The temperature value displayed on the upper side of the screen position is the one measured by the device predefined in the configuration section, that can be considered as the water temperature measurement.

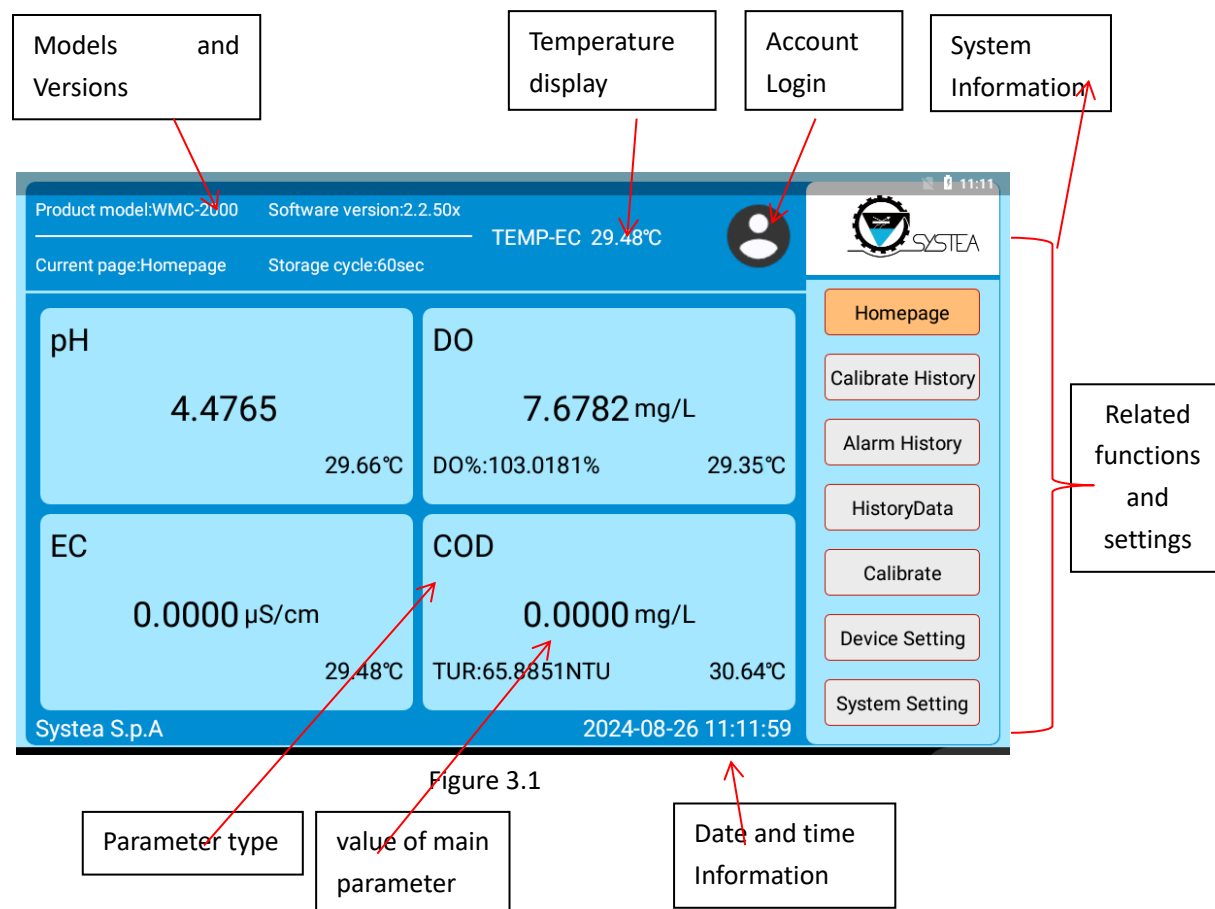


Figure 3.1

Clicking on the company logo in the upper right corner of the main interface, the product information and company information of the current controller are reported, as shown in the following figure.

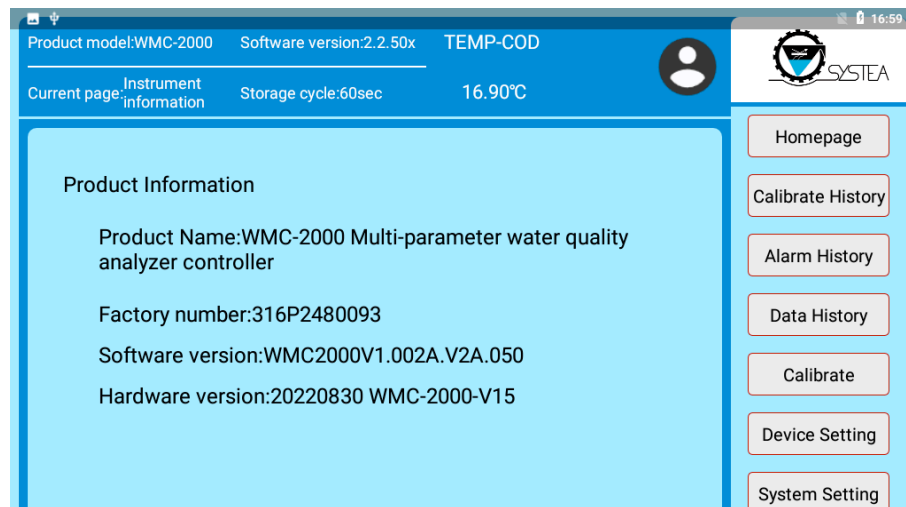


Figure 3.2

3.2 User login

When you click the Login button in the main interface, the interface shown in the figure below will pop up, enter the account number "user" and the password "123456" and click Login to complete the login.

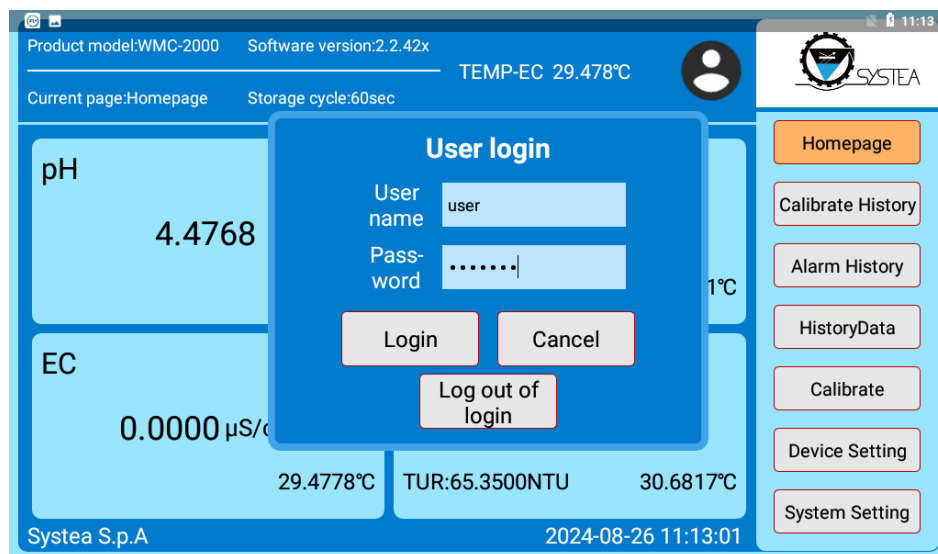


Figure 3.3 User login

3.3 Calibrate history

Click "**Calibrate History**" in the main interface to enter the calibration record query interface, click **Search** button to query all the calibration information of each device connected to the controller. The content includes Port1-address, device type, calibration content (calibration point) and calibration time.

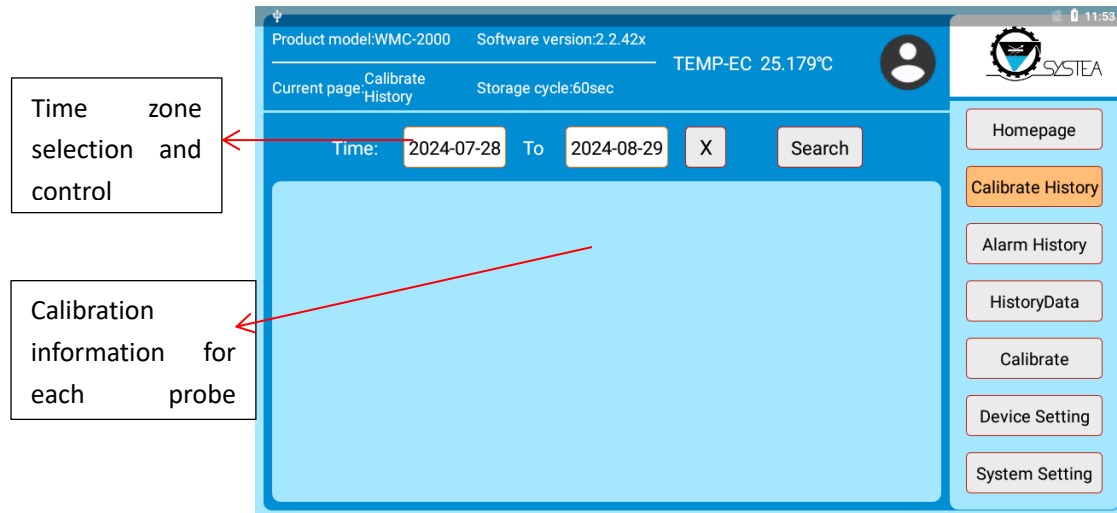


Figure 3.4 Calibration history

3.4 Alarm history

Click "**Alarm History**" in the main interface to enter the alarm record interface, and then click **Search** button to query the abnormal alarm information of each device connected to the controller. The content includes port1-address, device type, alarm content and alarm time.

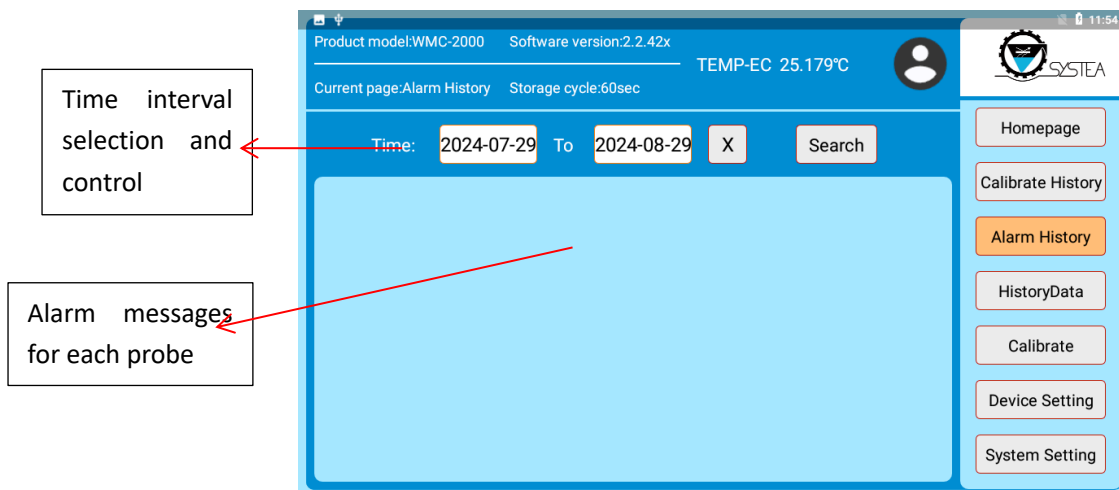


Figure 3.5 Alarm history

3.5 History data reading

To query the historical data of a device, click the **"HistoryData"** button, select the device in the displayed pop-up window (Fig.3.6) and the query interval, that can be selected between minute, every 10 minutes, every 30 minutes, every hour, then the query can be completed by clicking the **"Query"** button. After waiting for the data reading (may need to buffer for a period of time), the interface shown in the figure below will pop up, and you can slide up and down (Fig.3.7) to find the date relative to the selected time period. After reading the data, you can input the corresponding number of pages to jump to and then click jump to search the related pages.



Figure 3.6

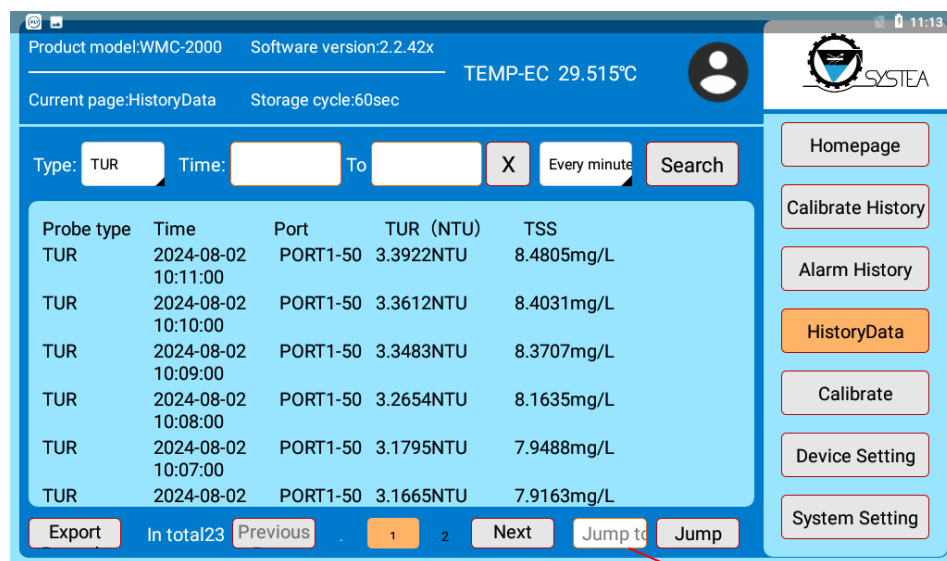


Figure 3.7

Enter the number of pages to jump to

Using the query section, it is possible to select the parameter and the data range between two times, then pushing the query button the results will be displayed.

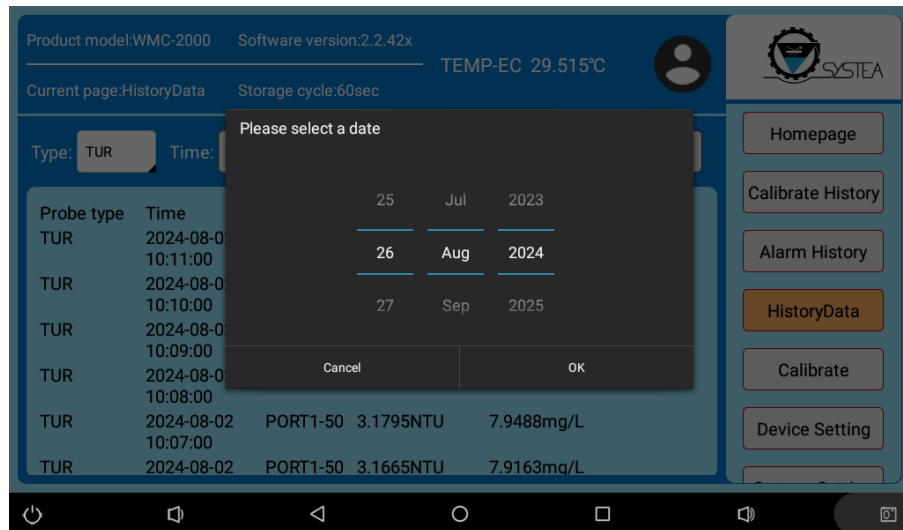


Figure 3.8

3.6 Calibration

Click the "Calibrate" interface button in the main interface to enter the calibration interface as shown below.

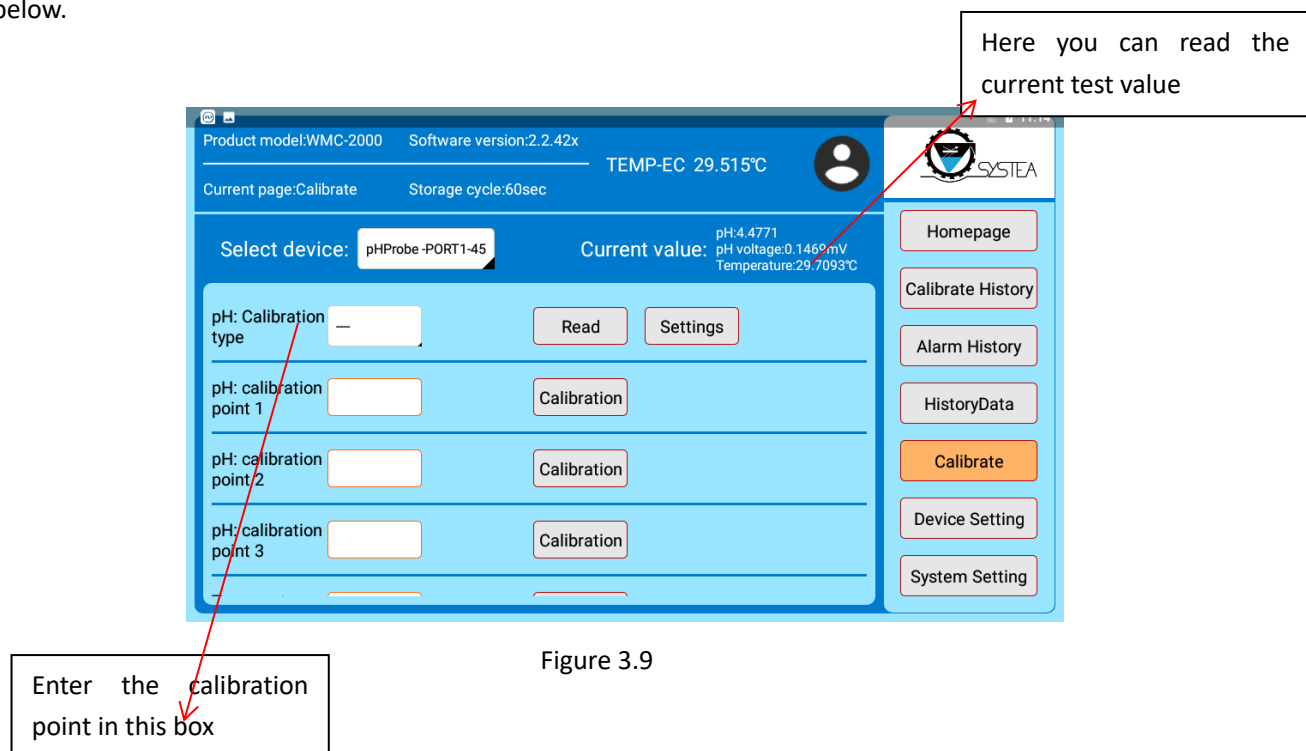


Figure 3.9

The operation of the calibration is based on the description of the specific device: keeping pH sensor as an example, please select the pH device in the selection of equipment window (ref. Fig. 3.6) and the pH calibration interface will be displayed (Fig.3.9). The current measurement of the pH value information is displayed in the "current value" section: please check it, it to make sure the value remains stable during the calibration sequence. In case of sensors requiring zero calibration, please enter the value 0 in the Zero

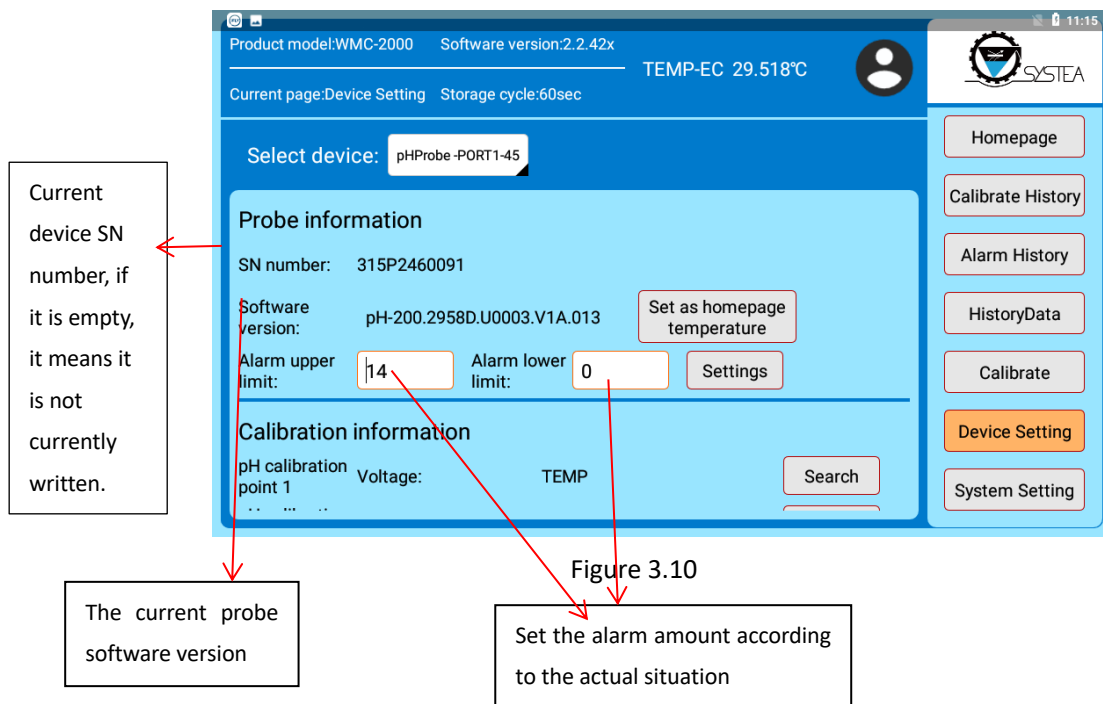
calibration setup box and press the "Settings" button. The calibration dialogue box will pop-up, prompting for the calibration, so click "Confirm. Then the message "calibration process takes 60 seconds, please wait patiently" will be displayed; at the end of the system calibration, the calibration dialogue box will pop up informing the success of calibration. In the upper right corner, the current value and the calibration value will be the same and this will confirm the end of the calibration procedure.

Note that after the interface prompts the calibration is complete, it is important to wait until the current measured value will fit with the value expected for the calibration.

3.7 Device setting

3.7.1 Device information

Click "Device Setting" in the main interface to enter the probe information setting interface. Complete the setting of upper and lower alarm limits and the query of equipment SN number and software version, in which, if the SN number is not written, the corresponding position will be blank. The upper and lower alarm limits are determined according to the requirements of the equipment connected to the current port, to determine the value of the upper and lower limits of the alarm for water sample anomalies, so enter the limit value directly into the corresponding input box and then click on the right side of the "Settings" button to complete the setup.



3.7.2 Calibration information

In the window shown below (Fig.3.11), you can read the pH probe "calibration information", click on the query to read the most recent calibration of the calibration point "time, calibration value, voltage". As shown in the figure below:

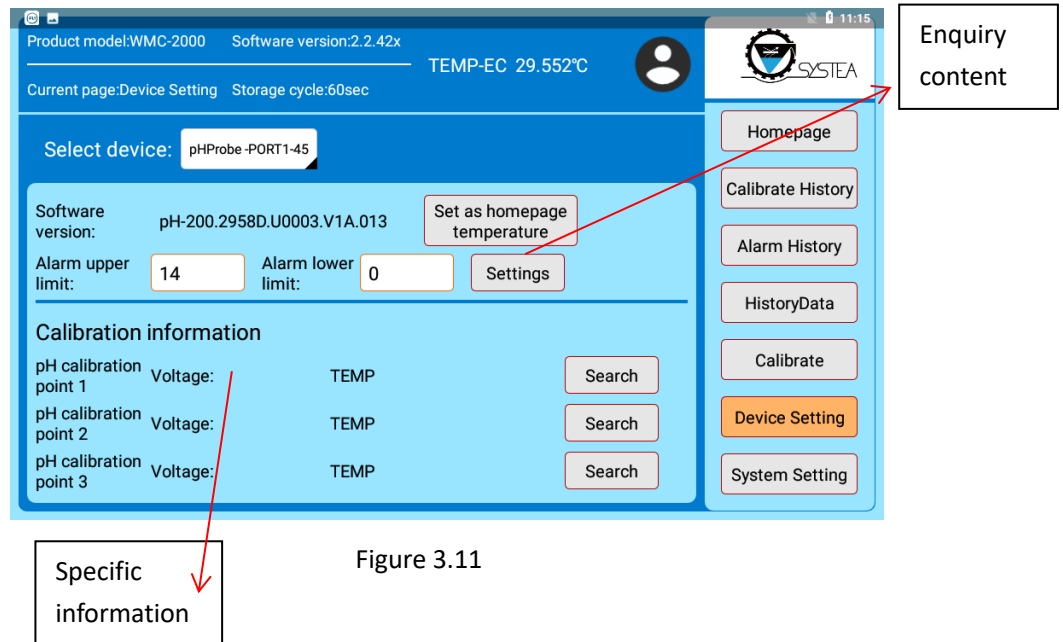


Figure 3.11

3.8 System setting

3.8.1 Data setup

You can enter the system setting interface by clicking the "System Setting" button in the main interface, and the interface after entering is shown below. In this interface, you can set the interval of saving historical data, the interval of uploading data and the number of decimal digits to be displayed for each selected parameter. According to the requirement, you can directly input the desired interval value in the input box, and click "Set" on the right side to complete the setting.

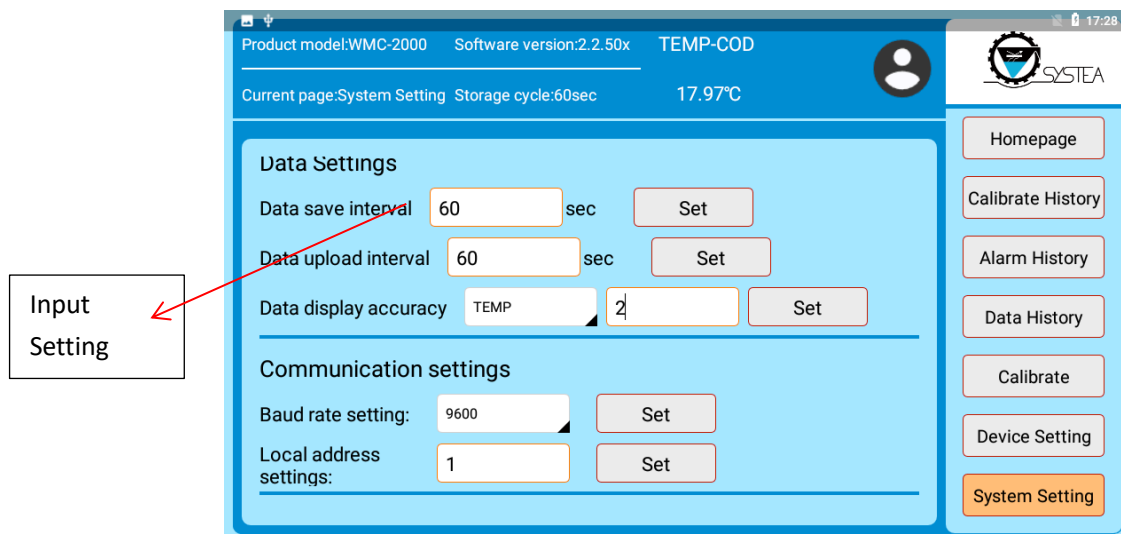


Figure 3.12

Note: The setting interval needs to be an integer multiple of 60.

3.8.2 Screen Brightness and Sleep settings

In the "Communication Settings" interface, scroll down to the "Screen Brightness and Sleep Settings" interface.

1) Default Brightness Adjustment: Input the desired brightness value (within the range of 13-100) in the Default Brightness Adjustment field, and then click on "Settings" to write the settings for the Default Brightness

2) Sleep Brightness Settings: Input the desired brightness value (within 13-100 range) in the Default Brightness Adjustment column, click "Settings" to write the setting for Default Brightness.

3) Sleep Time Settings: Write the desired hibernation time (within 10-42000) in the Sleep Time column, click "Set" to write the setting for Sleep Time.

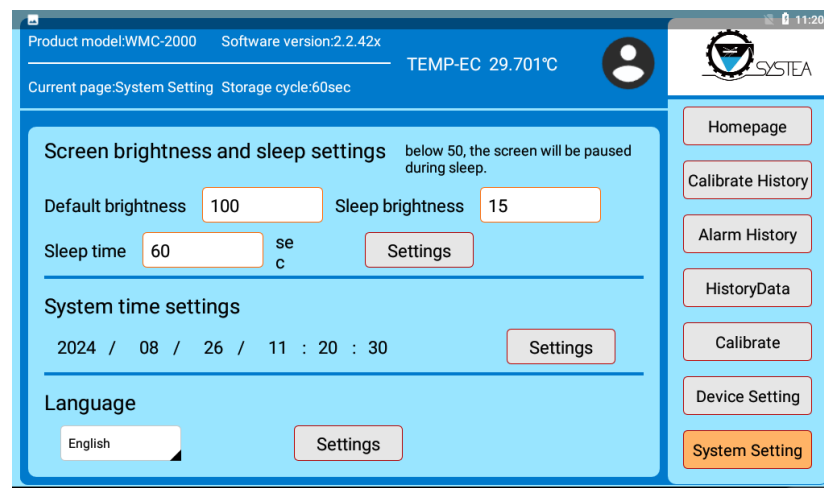


Figure 3.13

3.8.3 System time

In the system time column, you can read the current time directly. When modifying the system time, click "Settings" to bring up the date and time setting interface, and select Set Date and Set Time to modify the system time respectively.

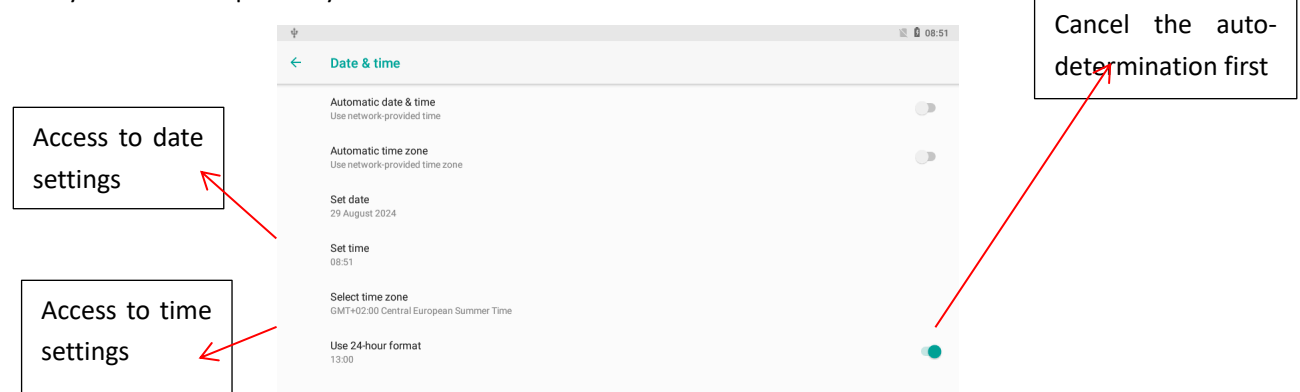


Figure 3.14

3.8.4 Analog outputs (optional)

WMC2000 controller can be equipped by optional HW modules, shown in Figure 3.15:



Figure 3.15 External analog outputs HW modules

to generate 4-20 mA analog outputs for up to five channels; the available options are the following:

- SH-4x420 - Set n.2 Modbus 485 external modules to generate n.4 4-20 mA
- SH-2x420 - Modbus 485 external module to generate additional one 4-20 mA.

The indicative dimensions of each HW module are the following: 12.5 (H) x 7.5 (W) x 3 (D) cm.

The Modbus external HW modules have to be connected to the same Modbus 485 serial port.

In the following figure, the analog output settings window is shown.

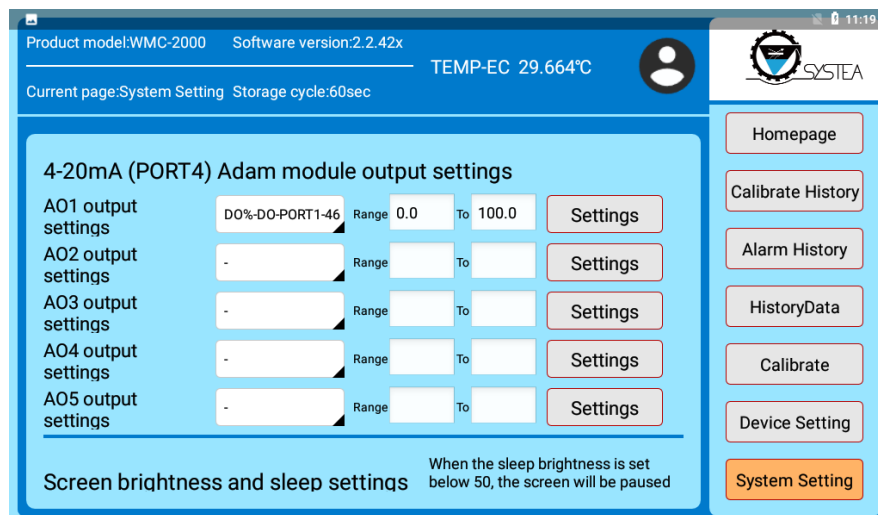


Figure 3.16 4-20 analog outputs settings

3.8.5 Digital outputs (optional)

WMC2000 controller can be equipped with one or two optional HW modules, shown in Figure 3.16:



Figure 3.17 External digital output HW modules

They have to be connected to the same Modbus 485 serial port as above, to generate n.6+7 digital outputs as described here below:

Module n.1:

- ch1-4: Alarm signals
- ch5: Calibration control 1
- ch6: Automatic cleaning control
- ch7: Empty

Module n.2:

- ch1-3: Calibration control 2-4
- ch4-7: Threshold concentration overlap

In the following figure, the related configuration window is shown.

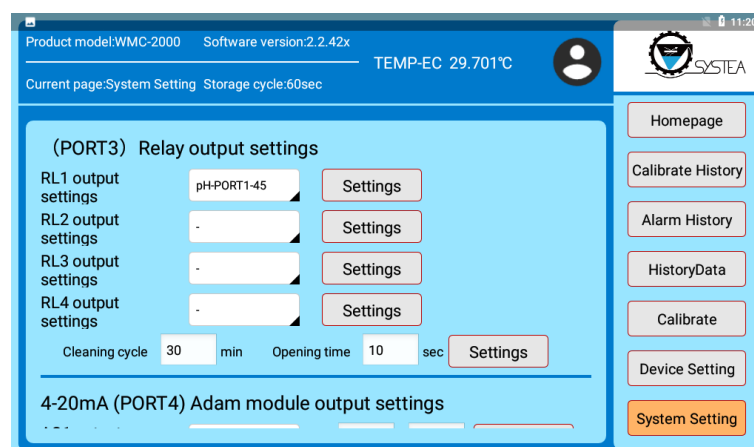


Figure 3.18 Relay output settings

4 Cautions

4.1 Storage and Custody

When the instrument you purchased is not to be installed urgently or needs to be put aside for a longer period of time, please store it properly and follow the below listed requirements:

- ambient temperature below -20°C or above 60°C;
- direct sunlight;
- next to the heater;
- vibrating or dusty environment;
- humid environment.

4.2 Precautions in use

Please avoid to use the controller in a humid environment or in a place that is susceptible to splashing water.

4.3 Troubleshooting

No.	Fault type	Possible causes	Suggestion
1	Interface is stucked	System crash	Please wait some time, then try to use it again; if the problem is not solved, please turn OFF, wait a few seconds and then turn ON again the device. If even if after rebooting the issue persists, the device should be returned for factory maintenance
		Longer data reading time	
		Equipment failure	
2	Communication anomalies	Analytical instrument communication abnormality	When the communication is abnormal, first of all, you can switch port port to check whether the analytical instrument communication is abnormal; then you can try to reconnect the line to test whether the contact is poor; if still can not be solved need to return to the factory for maintenance
		Poor wire connection contact	
		Controller damaged	

Table 4.1 Typical faults and handling

4.4 Unpacking and inspection

When you receive the purchased instrument to open the box before the inspection, please carefully check the packaging for any damage or signs of falling, if there is any suspicion and cause damage to the contents of the goods, quickly notify the company and the carrier! Please be careful when opening the box, do not damage the instrument or other parts. After opening the box, please check each piece, if you find the wrong shipment, missing or damaged, please notify the company as soon as possible.

4.5 After sale service

If during the installation, commissioning and use of the device any problem would raise, please contact our company in a timely manner, especially if a blocking type fault would be found. When reporting back to our company, please inform about the instrument number, model number, date of purchase, including a clear description of the failure, to facilitate our quick and accurate technical support.

5 Modbus RTU communication protocol

5.1 Types and formats of communication data:

The message transfer is asynchronous and is in bytes. The communication messages passed between master and slave are in 11-bit word format is shown in the following table:

Word format (serial data)	11-bit binary
starting position	1 place
data bit	8-bit
parity bit	not have
stop bit	1 bit (1 bit used by default)

Table 5.1.1

5.2 Communication data (message frame) format

address field	function code	data area	CRC check
1 byte	1 byte	Maximum 252 bytes	16-bit CRC code (redundant cyclic code)

Table 5.2.1

5.3 Communication message transfer process

When a communication command is sent from the sending device (host) to the receiving device (slave), the slave that meets the corresponding address code receives the communication command and reads the information according to the function code and the relevant requirements, and then performs the corresponding task if the CRC checksum is correct, and then returns the execution result (data) to the host. The returned information includes address code, function code, data after execution and CRC check code. If the CRC check is wrong, no information is returned.

5.3.1 Address Code

The address code is the first byte (8 bits) of each communication message frame, from 0 to 254. This byte indicates that the slave whose address is set by the user will receive the message sent by the master. Each slave must have a unique address code, and only slaves that match the address code can respond to a return message. When a slave sends back a message, the return data begins with the respective address code. The address code sent by the host indicates the address of the slave to which it will be sent, and the address code returned by the slave indicates the address of the slave to which it is being sent back. The corresponding address code indicates where the message came from.

5.3.2 Function Code

The function code is the second byte transmitted in each communication message frame; the ModBus communication protocol can define function codes from 1 to 127:

- Sent as a request from the host, the function code tells the slave what action to perform;
- Sent as a response from the slave, the function code returned by the slave is the same as the one sent from the host and it indicates that the slave has responded to the host and it has performed

the relevant action.

function code	definition	Operation (Binary)
0x03	Read Holding Register Data	Read data from one or more registers
0x04	Read Input Register	Reads data from one or more input registers
0x06	Write Single Holding Register	Write a single set of data to a single register
0x10	Write Multiple Holding Registers	Write multiple sets of binary data to multiple registers

Table 5.1 - MODBUS partial function codes

5.3.3 Data area

The data area includes what information needs to be returned or what action needs to be performed by the slave. This information can be data (e.g., switching inputs/outputs, analogue inputs/outputs, registers, etc.), reference addresses, etc. For example, if the host tells the slave to return the value of a register (including the starting address of the register to be read and the length of the register to be read) via function code 03, the returned data includes the length of the register data and the data content. For different slaves, the address and data information are different (communication information table should be given).

Multi-meter controller adopts Modbus communication protocol, the host computer (PLC, RTU, PC, DCS, etc.) can read its data register at will by using communication commands (function code 03) (see appendix for details of its data information table).

The format of the commands responded to by the multi-meter header controller is from the machine address, function code, data area, and CRC code. The data in the data area are all two bytes and are preceded by the high bit.

5.3.4 Standstill time requirements

Sending data requires that the interval between two frames of data has to be greater than 3.5 characters of transmission time.

5.3.5 Register Definition

- ① Holding registers: 0x0000-0x0016, 0x0102-0x116.

5.4 MODBUS Function Codes

5.4.1 Function code "03": read multiplexed holding register inputs

As an example, the master wants to read the data of 2 slave registers with address 01 and start address 2000. The address and data of the slave data registers are:

register address	Register data (hexadecimal)
2000	0E10
2001	C0CB

The format of the message sent by the host:

Host sends	byte count	Message sent	Remarks
slave address	1	01	Send to slave at address 01
function code	1	03	registers
starting address	2	2000	Starting address 2000
Read Data Length	2	0002	Read 2 registers (4 bytes total)
CRC code	2	CFCB	The CRC code is calculated by the host

The format of the message returned by the slave:

slave response	byte count	Information returned	Remarks
slave address	1	01	From Slave 01
function code	1	03	registers
Data length (number of bytes)	1	08	Total 8 bytes
Data in register 1	2	0E10	The contents of register 2000 at address
Data in register 2	2	C0CB	The contents of register 2001 at address
CRC code	2	6089	The CRC code is calculated by the slave

5.4.2 Function code "04": read multi-input registers

For example, the master wants to read the data of 2 slave registers with address 01 and start address 2000. The address and data of the slave data registers are:

register address	Register data (hexadecimal)	corresponding parameter
2000	0E10	
2001	C0CB	

The format of the message sent by the host:

Host sends	byte count	Message sent	Remarks
slave address	1	01	Send to slave at address 01
function code	1	04	registers
starting address	2	2000	Starting address 2000
Read Data Length	2	0002	Read 2 registers (4 bytes total)
CRC code	2	7A0B	The CRC code is calculated by the host

The format of the message returned by the slave response:

slave response	byte count	Information returned	Remarks
slave address	1	01	From Slave 01
function code	1	04	registers
Data length (number of bytes)	1	04	Total 4 bytes
Data in register 1	2	0E10	The contents of register 2000 at address
Data in register 2	2	C0CB	The contents of register 2001 at address
CRC code	2	E93E	The CRC code is calculated by the slave

5.4.3 Function code "10": Write Multiple Holding Registers

The host uses this function code to save multiple data into the MM-500's data memory. registers in the Modbus communication protocol refer to 16 bits (i.e., 2 bytes) and are preceded by the high bit.

For example, the master wants to save 0E10 to the slave register at address 2000 (slave address code 01).

The format of the message sent by the host:

Host sends	byte count	Message sent	Remarks
slave address	1	01	Send to slave at address 01
function code	1	10	Write multiplexed registers
starting address	2	2000	Starting address of the register to be written to
Save data length	2	0001	Word length of saved data
Save data byte length	1	02	Byte length of saved data (4 bytes)
Save Data 1	2	0E10	Data to be written to address 2000
CRC code	2	823E	The CRC code is calculated by the host

The format of the message returned by the slave response:

slave response	byte count	Message sent	Remarks
slave address	1	01	Send to slave at address 01
function code	1	10	Write multiplexed registers
starting address	2	2000	Starting address of the register to be written to
Save data length	2	0001	Word length of saved data
CRC code	2	0A09	The CRC code is calculated by the slave

5.5 Error Checksum (CRC checksum)

The host or the slave can use the check code to judge whether the received information is correct or not. Due to electronic noise or some other interference, information in the transmission process sometimes occurs in error, error check code (CRC) can check the host or slave in the process of communication data transmission information whether there is an error, the wrong data can be abandoned (either send or receive), so as to increase the safety and efficiency of the system.

The CRC (Cyclic Redundancy Code) of the MODBUS communication protocol consists of 2 bytes, i.e. a 16-bit binary number. The CRC code is sent by the sending device.

(host) is calculated and placed at the end of the frame in which the information was sent. The device receiving the message (slave) then recalculates the CRC of the received message, compares whether the calculated CRC matches the received one, and if they do not match, an error is indicated.

5.5.1 CRC code calculation method

- 1 Preset a 16-bit register to hex FFFF (i.e., all 1s); call this register the CRC register;
- 2 Isolate the first 8-bit binary data (the first byte of the communication information frame) with the lower 8 bits of the 16-bit CRC register, and put the result in the CRC register;
- 3 Shift the contents of the CRC register right by one bit (towards the lower bit) to fill the highest bit with 0, and check the shifted-out bit after the right shift;
- 4 If the shift out bit is 0: Repeat step 3 (shift right one bit again); If the shift out bit is 1: CRC register with polynomial A001;
- 5 (1010 0000 0000 0001) for dissimilarity;
- 6 repeat steps 3 and 4, until the right shift 8 times, so that the entire 8-bit data are all processed;
- 7 Repeat steps 2 to 5 for the next byte of the communication information frame;
- 8 Exchange the high and low bytes of the 16-bit CRC register obtained after all bytes of this communication information frame have been calculated according to the above steps;
- 9 The content of the CRC register obtained at last is: CRC code.

C routines are listed below:

```
unsigned short crc_calc(unsigned char *p,unsigned char len)
{
    //1. Prepare CRC checksum initial value.
    unsigned short crc_data =
    0xffff; unsigned char i ;
    unsigned char j ;
    //6. Repeat steps 2 to 5 for the next byte of the communication
    message frame; while( len-- )
    {
        i = *p++;
        //2, lower 8 bits of CRC
        register      phase
        dissimilarity  crc_data
        ^= i;
```

```
//5. Repeat steps 3 and 4 until it is shifted right 8 times, so that the
whole 8-bit data are all processed; for(j=0;j<8;j++)
{
    //4.1, If the shift out bit is 1: the CRC register is differentiated with the polynomial A001 (1010 0000 0000
0001);
    if( crc_data & 0x01 )
    {
        //3. Shift the content of the CRC register right one bit (towards the lower bit) to fill the highest bit
with 0 and check the shifted out bit after the right shift;
        crc_data >>= 1;

        crc_data ^= 0xA001.
    }
    //4.2. If the shifted out bit is 0: repeat step 3 (shift right one bit again);
    else
    {
        crc_data >>= 1;
    }
}

}

//7. Swap the high and low bytes of the 16-bit CRC register obtained after all bytes of this communication information
frame have been calculated according to the above steps;
i = crc_data >> 8;
crc_data <<= 8; crc_data
|= i;
//8, the last obtained CRC register content is: CRC
code. return crc_data.
}
```

5.6 Alarm codes lists (expressed in bits)

Modbus-RTU mode, serial port default parameter configuration:

baud	9600
data bit	8
stop bit	1
parity check	None
slave address	0x01

Modbus-RTU mode, data transfer size end description:

data type	Big and small end model
Float (encoding format IEEE754)	the main points

5.6.1 SH-ISENH4200 Ammonia probe

Alarm code	Reason for failure
1ul<<1	Temperature difference greater than 5°C during pH calibration
1ul<<2	Abnormal slope for pH calibration
1ul<<4	Ammonia nitrogen calibration with a temperature difference of more than 5°C
1ul<<5	Abnormal slope during ammonia calibration
1ul<<6	Ammonia nitrogen concentration greater than set concentration
1ul<<8	Temperature less than 0 or more than 50
1ul<<9	Failure to calibrate actual water samples
1ul<<14	Ammonia nitrogen over range

5.6.2 SH-TUR200 Turbidity sensor

Alarm code	Reason for failure
1ul<<0	Turbidity over range
1ul<<1	Motor failure
1ul<<2	LED light source alarm
1ul<<3	Calibration failure
1ul<<4	Hall Failure
1ul<<14	Turbidity above range

5.6.3 SH-DO200 Dissolved oxygen sensor

Alarm code	Reason for failure
1<<0	Abnormal light intensity signals
1<<2	Temperature measurement abnormality
1<<3	Measured signal abnormality
1<<5	Zero calibration failure
1ul<<14	Dissolved oxygen exceeds the range

5.6.4 SH-pH200 pH sensor

Alarm code	Reason for failure
1ul<<0	Abnormal pH signal
1ul<<1	Excessive temperature difference between calibration points
1ul<<2	Calibration slope anomaly
1ul<<3	Excessive temperature difference between calibration points

1ul<<4	Calibration slope anomaly
1ul<<8	Temperature measurement abnormality
1ul<<14	pH over range

5.6.5 SH-UVCOD200 COD probe

Alarm code	Reason for failure
1ul<<0	Motor failure, gear slippage
1ul<<1	Motor failure, blockage
1ul<<14	COD out of range

5.6.6 SH-EC200 Conductivity sensor

Alarm code	Reason for failure
1ul<<0	Temperature less than 0 Greater than 50
1ul<<14	Conductivity over range

5.6.7 SH-CHL200 Chlorophyll sensor

Alarm code	Reason for failure
1ul<<0	Chlorophyll over range
1ul<<1	motor failure
1ul<<2	Temperature over range
1ul<<14	Chlorophyll over range

5.6.8 SH-ORP200 ORP sensor

Alarm code	Reason for failure
1ul<<8	Temperature out of range
1ul<<9	ORP Over Range
1ul<<14	ORP out of range

5.6.9 SH-BGA200 Blue-Green Algae sensor

Alarm code	Reason for failure
1ul<<0	Blue-green algae overrange
1ul<<1	motor failure
1ul<<2	Temperature over range
1ul<<14	Blue-green algae exceeding the range

5.6.10 SH-ISENO3 Nitrate probe

Alarm code	Reason for failure
1ul<<0	Abnormal pH signal
1ul<<1	pH calibration for temperature differences greater than 5°C
1ul<<2	Abnormal slope for pH calibration
1ul<<3	Abnormal nitrate-nitrogen signaling
1ul<<4	Nitrate and nitrogen calibration with temperature difference more than 5°C.
1ul<<5	Abnormal slope in nitrate-nitrogen calibration
1ul<<6	Nitrate-nitrogen concentration greater than set concentration
1ul<<7	pH greater than 12
1ul<<8	Temperature less than 0 or more than 50
1ul<<14	Nitrate nitrogen exceeding the range

5.6.11 SH-FCI200 Residual chlorine measurement device

Alarm code	Reason for failure
1ul<<0	Abnormal pH signal
1ul<<1	pH calibration with too large a temperature difference
1ul<<2	Abnormal slope for pH calibration
1ul<<3	pH concentration greater than 9 or less than 6
1ul<<4	Temperature out of range
1ul<<5	Residual chlorine voltage over range
1ul<<14	Chlorine residual above range

5.6.12 SH-TDS Total dissolved solids sensor

Alarm code	Reason for failure
1ul<<0	Temperature less than 0 Greater than 50
1ul<<14	TDS over range

5.6.13 SH-ISEF200 Fluoride probe

Alarm code	Reason for failure
1ul<<1	Temperature difference greater than 5° C during pH calibration
1ul<<2	Abnormal slope during pH calibration
1ul<<4	Fluoride ion calibration with a temperature difference of more than 5° C
1ul<<5	Abnormal slope during fluoride ion calibration
1ul<<6	Fluoride ion concentration greater than set concentration
1ul<<7	pH concentration greater than 12
1ul<<8	Temperature less than 0 or more than 50
1ul<<9	Failure to calibrate actual water samples
1ul<<14	Fluoride ions exceeding the range

5.6.14 SH-ISCIF200 Chloride probe

Alarm code	Reason for failure
1ul<<1	Temperature difference greater than 5° C during pH calibration
1ul<<2	Abnormal slope during pH calibration
1ul<<4	Chloride ion calibration with a temperature difference of more than 5° C
1ul<<5	Abnormal slope during chloride ion calibration
1ul<<8	Temperature less than 0 or more than 50
1ul<<14	Chlorine ions exceeding the range

5.6.15 SH-MLSS200 Sludge sensor

Alarm code	Reason for failure
1ul<<0	Signal overrange
1ul<<1	motor failure
1ul<<2	LED light source alarm
1ul<<4	Hall Failure
1ul<<14	Sludge concentration exceeding the measuring range

5.6.16 SH-SS200 Suspended solids sensor

Alarm code	Reason for failure
1ul<<0	Signal overrange
1ul<<1	motor failure
1ul<<2	LED light source alarm
1ul<<4	Hall Failure
1ul<<14	Suspended matter exceeding the measuring range

5.6.17 SH-FT200 Ultra-low turbidity measurement device

fault location	Reason for failure
1ul<<14	Low turbidity exceeds the range

5.7 Holding registers

5.7.1 Parameters

Parameter	Typology	Meas. unit	Range of values	Bytes	Input register address	Note
Temperature	Float	°C		4	0x0000	
Ammonia	Float	mg/L		4	0x0002	--
Turbidity	Float	NTU		4	0x0004	
Dissolved oxygen	Float	mg/L		4	0x0006	
pH	Float	pH		4	0x0008	
COD	Float	mg/L		4	0x000a	
Conductivity	Float	uS/cm		4	0x000c	
Chlorophyll	Float	ug/L		4	0x000e	
ORP	Float	mV		4	0x0010	
Cyanobacteria (Blue-Green Algae)	Float	cells/mL		4	0x0012	
Nitrate	Float	mg/L		4	0x0014	
Residual chlorine	Float	mg/L		4	0x0016	

TDS	Float	mg/L		4	0x0018	
Fluoride	Float	mg/L		4	0x001a	
Chloride	Float	mg/L		4	0x001c	
Sludge concentration	Float	mg/L		4	0x001e	
Suspended solids	Float	mg/L		4	0x0020	
Ultra-low turbidity	Float	NTU		4	0x0022	

Table 5.7 Holding registers – parameters table

5.7.2 Alarms

Parameter	Typology	Meas. unit	Range of values	Bytes	Input register address	Note
Ammonia alarm code	uint32			4	0x0102	See alarm code table
Turbidity alarm code	uint32			4	0x0104	See alarm code table
Dissolved oxygen alarm code	uint32			4	0x0106	See alarm code table
pH alarm code	uint32			4	0x0108	See alarm code table
COD alarm code	uint32			4	0x010a	See alarm code table
Conductivity alarm code	uint32			4	0x010c	See alarm code table
Chlorophyll alarm code	uint32			4	0x010e	See alarm code table

ORP alarm code	uint32			4	0x0110	See alarm code table
Cyanobacteria (BGA) alarm code	uint32			4	0x0112	See alarm code table
Nitrate alarm code	uint32			4	0x0114	See alarm code table
Residual chlorine alarm code	uint32			4	0x0116	See alarm code table
TDS alarm code	uint32			4	0x0118	See alarm code table
Fluoride alarm code	uint32			4	0x011a	See alarm code table
Chloride alarm code	uint32			4	0x011c	See alarm code table
Sludge concentration alarm code	uint32			4	0x011e	See alarm code table
TSS alarm code	uint32			4	0x0120	See alarm code table
Ultra-low turbidity alarm code	uint32			4	0x0122	See alarm code table

Table 5.8 Holding registers – parameters alarms table

5.7.3 Status

Parameter	Typology	Meas. unit	Range of values	Bytes	Input register address	Note
Ammonia status	uint16			2	0x0202	0: Measurement 1: pH1 is being calibrated 2: pH2 is being calibrated 3: pH3 is being calibrated 4: Ammonia 1 is being calibrated 5: Ammonia 2 is being calibrated
Turbidity status	uint16			2	0x0204	1: Measurement 2: Being calibrated 3: Motor operation
Dissolved oxygen status	uint16			2	0x0206	0: Measurement 1: Calibration
pH status	uint16			2	0x0208	0: Measurement Three-point calibration 1: pH1 is being calibrated 2: pH2 is being calibrated 3: pH3 is being calibrated Two-point calibration 4: pH1 is being calibrated 5: pH2 is being calibrated
COD status	uint16			2	0x020a	0: Idle 1: Measurement 2: Being calibrated

						3: Motor operation
Conductivity status	uint16			2	0x020c	0: Measurement 1: Calibration in progress
Chlorophyll status	uint16			2	0x020e	1: Measurement 2: Motor stops waiting for solution to stabilise 3: Motor operation 4: Find the motor stop voltage 5: Motor stop timeout 6: Calibration first point (low range) 7: Calibration second point (low range) 8: Calibration of the first point (high range) 9: Calibration of the second point (high range)
ORP status	uint16			2	0x0210	0: Measurement
Cyanobacteria (BGA) status	uint16			2	0x0212	1: Measurement 3: Motor operation 4: Find the motor stop voltage 5: Motor stop timeout 6: Motor stops waiting for solution to stabilise 8: Calibration of the first point 9: Calibration of the second point

Nitrate status	uint16			2	0x0214	0: Measurement 1: pH1 is being calibrated 2: pH2 is being calibrated 3: pH3 is being calibrated 4: Nitrate-Nitrogen 1 is being calibrated 5: Nitrogen 2 is being calibrated
Residual chlorine Status	uint16			2	0x0216	0: Measurement
TDS status	uint16			2	0x0218	0: Measurement 1: Calibration in progress
Fluoride status	uint16			2	0x021a	0: Measurement 1: pH1 is being calibrated 2: pH2 is being calibrated 3: pH3 is being calibrated 4: Fluoride ion 1 is being calibrated 5: Fluoride ion 2 is being calibrated
Chloride status	uint16			2	0x021c	0: Measurement 1: pH1 is being calibrated 2: pH2 is being calibrated 3: pH3 is being calibrated 4: Chloride 1 is being calibrated 5: Chloride 2 is being calibrated
Sludge concentration status	uint16			2	0x021e	1: Measurement 2: Being calibrated

						3: Motor operation
TSS status	uint16			2	0x0220	1: Measurement 2: Being calibrated 3: Motor operation
Ultra-low turbidity status	uint16			2	0x0218	0: Measurement 1: Calibration in progress

Table 5.9 Holding registers – parameters status table

5.7.4 Subparameters

Parameter	Typology	Meas. unit	Range of values	Bytes	Input register address	Note
Ammonia subparameter (pH)	Float	pH		4	0x0052	--
COD subparameter (turbidity)	Float	NTU		4	0x005a	
Nitrate subparameter (pH)	Float	pH		4	0x0064	
Residual chlorine subparameter (pH)	Float	mg/L		4	0x0066	
Fluoride subparameter (pH)	Float	mg/L		4	0x006a	
Chloride subparameter (pH)	Float	mg/L		4	0x006c	

Table 5.7 Holding registers – subparameters table

5.8 MODBUS-RTU protocol example (address default 0x01)

5.8.1 Read a single parameter

Temperature for example

Send: 01 03 00 00 00 00 02 c4 0b

01: device address, 03: read holding register, 0000: register start address, 0002: read 2 registers, c40b: check digit

Received: 01 03 04 40 d3 1a a0 15 12

01: device address, 03: read hold register, 04: 4 bytes, 40d31aa0 (6.597 big end): master parameter, 1512: checksum

5.8.2 Read alarm code

Chlorophyll for example

Send: 01 03 01 0e 00 02 a4 34

01: device address, 03: read holding register, 010e: register start address, 0002: read 2 registers, a434: check digit

Received: 01 03 04 00 00 40 00 cb f3

01: device address, 03: read hold register, 04: 4 bytes, 00004000 (converted to binary 0100 0000 0000 0000): chlorophyll out of range, cbf3: check digit